

ABSTRACT

AN INTELLIGENT WORKSPACE FOR DEVELOPERS AND TEAMS

TaskNest is a web-based, agentic AI-powered project management platform designed to solve the fragmentation problem faced by developers and student teams managing multiple projects simultaneously. It provides a centralized, intelligent workspace where users can organize projects into dedicated folders, each containing its own isolated knowledge base, notes section, and task management system.

The platform integrates Retrieval-Augmented Generation (RAG) technology with a multi-agent system orchestrated through LangGraph, enabling a context-aware AI chatbot that adapts its knowledge base dynamically based on the active project folder. Each project folder contains a Resources section where users can upload documents such as PDFs and DOCX files. These resources are automatically ingested, chunked, embedded using state-of-the-art embedding models, and stored in a project-specific ChromaDB vector collection. This ensures that the RAG brain is completely isolated per project — when a user navigates to a different folder, the chatbot's knowledge switches entirely to that folder's resources.

The system's AI orchestrator, powered by Groq's llama3-70b model and managed through LangGraph's StateGraph, routes user queries to specialized agents: a Folder Agent for project management, a Resource Agent for document handling, a Notes Agent for intelligent note-taking, and a Task Agent for todo management. Users can interact with the chatbot using natural language to perform any platform operation automatically, eliminating manual navigation.

A standout feature is the "Copy to Notes" functionality, which allows

users to save any chatbot response directly to the project's notes section with a single click. The notes can subsequently be edited, organized, and reviewed. The platform also features a Todo module per project, supporting task creation, status tracking, and deadline management.

Built on a modern stack — React.js with Vite and Tailwind CSS on the frontend, FastAPI and JWT authentication on the backend, Railway-hosted MySQL for relational data, and locally-run ChromaDB for vector storage — TaskNest delivers a seamless, intelligent, and scalable project management experience for students, developers, and engineering teams.